

Boolean Algebra and Circuit Simplification

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CIS 310: Computer Organization and Assembly Programming

Objectives

- ▶ Introduce SOP (Sum of Products) and POS (Product of Sums) representations.
- ▶ Understand the importance of circuit simplification.
- ▶ Learn techniques for Boolean expression simplification.
- ▶ Apply laws and rules of Boolean algebra to optimize circuits.
- ▶ Practice designing and simplifying digital circuits.

SOP and POS Representations

Definitions

- ▶ **SOP (Sum of Products):** A form where the expression is the sum (OR) of product (AND) terms.

Example: $f(A, B, C) = A \cdot B + A' \cdot C$

- ▶ **POS (Product of Sums):** A form where the expression is the product (AND) of sum (OR) terms.

Example: $f(A, B, C) = (A + B) \cdot (A' + C)$

Converting Truth Table to SOP/POS

1. Identify rows with output = 1 for SOP (or 0 for POS).
2. Form product terms for each row (SOP) or sum terms (POS).
3. Combine all terms using OR (SOP) or AND (POS).

SOP/POS Examples

Example Truth Table

A	B	C	Output
0	0	0	1
0	1	1	1
1	0	1	1
1	1	1	0

SOP Expression

$$\text{Output} = A'B'C' + A'BC + AB'C$$

POS Expression

$$(A + B + C)$$

Why Simplification?

Goals of Simplification

- ▶ Reduce the number of gates.
- ▶ Minimize fan-in and fan-out.
- ▶ Simplify interconnection complexity.
- ▶ Prevent circuit hazards.

Example Comparison

- ▶ **Unoptimized SOP Expression:**

$$f(A, B, C) = A \cdot B \cdot C + A \cdot B' \cdot C' + A' \cdot B \cdot C'$$

- ▶ **Optimized SOP Expression:**

$$f(A, B, C) = A \cdot B + A' \cdot C'$$

Simplification Techniques

Overview of Techniques

1. Apply Boolean Algebra Laws:
 - ▶ Commutative, Associative, Distributive Laws.
 - ▶ DeMorgan's Theorem.
2. Derive Minimum SOP or POS Forms.
3. Use Truth Tables to Cross-Verify.

Example

$$\text{Given: } f(A, B) = AB + AB' + A'B$$

$$\text{Simplified: } f(A, B) = A + B$$

Boolean Algebra Laws and Rules

Fundamental Laws

- ▶ **Commutative Law:** $A + B = B + A$, $A \cdot B = B \cdot A$
- ▶ **Associative Law:** $A + (B + C) = (A + B) + C$,
 $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
- ▶ **Distributive Law:** $A \cdot (B + C) = A \cdot B + A \cdot C$

DeMorgan's Theorem

$$\overline{A \cdot B} = \overline{A} + \overline{B}, \overline{A + B} = \overline{A} \cdot \overline{B}$$

Simplification Example

Problem

$$f(A, B, C) = A \cdot B \cdot C + A \cdot B' \cdot C + A \cdot B \cdot C'$$

Solution

1. Factorize common terms:

$$f(A, B, C) = A \cdot (B \cdot C + B' \cdot C + B \cdot C')$$

2. Simplify within parentheses:

$$f(A, B, C) = A \cdot (B + C)$$

Practice Slide 1: Simplify Using SOP Form

Problem: Simplify the following expression:

$$f(A, B, C) = A \cdot B' \cdot C + A' \cdot B \cdot C' + A \cdot B \cdot C$$

Steps:

1. Factorize common terms.
2. Apply Boolean algebra laws (Commutative, Associative, Distributive).
3. Write the final simplified SOP expression.

Practice Slide 2: Convert Truth Table to SOP/POS

Problem: Given the truth table:

A	B	C	Output
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

1. Write the SOP expression.
2. Write the POS expression.

Practical Application: Full Adder

Full Adder Boolean Expressions

- ▶ **Sum:** $S = (A \oplus B) \oplus C_{in}$
- ▶ **Carry:** $C_{out} = A \cdot B + C_{in} \cdot (A \oplus B)$

Circuit Design

- ▶ Design the circuit using AND, OR, and XOR gates.
- ▶ Simplify the design using SOP representation.

Conclusion and Next Steps

Key Takeaways

- ▶ SOP and POS provide structured representations of Boolean expressions.
- ▶ Simplification minimizes circuit complexity and cost.
- ▶ Boolean algebra laws are essential tools for optimization.

Next Lecture

- ▶ **Topic:** Karnaugh Maps and Visual Simplification Techniques.
- ▶ **Preparation:** Review SOP and POS conversions.